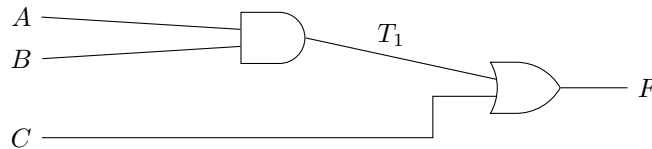


CSCI 160 Homework 5

- The logic diagram below has three inputs A , B , C and a single output F . Analyze the circuit: label the output of each intermediate gate, write the Boolean function F in terms of A , B , and C , and then build the complete truth table for F .



- A combinational circuit has three inputs x , y , z and two outputs F_1 and F_2 , described in words as follows:
 - $F_1 = 1$ whenever a *majority* of the inputs are 1 (that is, when two or more of x , y , z equal 1).
 - $F_2 = 1$ whenever *exactly one* of the inputs is 1.

For this circuit:

- Build the truth table for F_1 and F_2 .
 - From the truth table, write each output as a Boolean function (sum of minterms is fine, simplify if you can).
 - Draw the logic diagram using AND, OR, and NOT gates. Label your inputs and mark the outputs F_1 and F_2 .
- Build a 5-bit ripple-carry adder out of full adders and use it to add the two binary numbers

$$A = 10011 \quad \text{and} \quad B = 01011.$$

- Draw the block diagram of the 5-bit ripple-carry adder, showing the five full adders FA_0 through FA_4 , the inputs A_i and B_i , the carries C_0 through C_5 , and the sum outputs S_i . Take the input carry $C_0 = 0$.
 - Fill in a column-by-column table showing, for each position i , the carry-in C_i , the bits A_i and B_i , the sum bit S_i , and the carry-out C_{i+1} .
 - State the final 6-bit result $C_5S_4S_3S_2S_1S_0$ and check it by converting A , B , and the result to decimal.
- (BCD adder.)** Using the BCD addition method from class, add the two BCD digits

$$6 + 7$$

- Add the two digits in a 4-bit binary adder and write the raw binary sum $KZ_8Z_4Z_2Z_1$.
- Evaluate the correction condition $C = K + Z_8Z_4 + Z_8Z_2$ and state whether a correction is needed.

- (c) If a correction is needed, add 0110 and give the final answer as a carry digit and a corrected BCD digit. Confirm the decimal result.

5. **(Binary subtractor.)** A 4-bit adder-subtractor (the kind built from full adders with an XOR on each B input and a mode line M) is used to compute

$$A - B, \quad A = 1001 (= 9), \quad B = 0101 (= 5).$$

- (a) State the value of the mode bit M for subtraction, and give the value placed on the input carry C_0 .
(b) Form the 1's complement of B and then add $A + (1\text{'s comp of } B) + 1$ column by column.
(c) Give the 4-bit result. Explain what the end carry-out C_4 tells you in this case.

6. **(Multiplexer.)** Implement the Boolean function

$$F(A, B, C) = \sum m(0, 2, 4, 7)$$

using an 8-to-1 multiplexer.

- (a) Connect A, B, C to the selection lines $S_2S_1S_0$ (with A as the most significant). Write out the truth table for F .
(b) State what constant (0 or 1) each data input I_0 through I_7 should be tied to.
(c) Draw the multiplexer with its data inputs wired to the correct constants.